

Jawaharlal Nehru Govt. Engineering College, Sundernagar, Distt. Mandi (H.P)
Department of Computer Science & Engineering (AI&ML)

Program	B.Tech CSE	Semester	3rd CSE(AI&ML)
Subject Code	CSPC-313	Subject Title	CA&O
Mid Semester Test No.	2nd	Course Coordinator	Dr. Meenakshi Shruti Pal
Max. Marks	50	Time Duration	2 Hours
Date of MST	07.11.2024 (M)	Roll Number	

Note: Attempt all questions

Q. No.	Question	COs, RBT level	Marks
Q1	a. What do you understand by the hierarchy of memory? b. List the various characteristics of the multiprocessor. c. Discuss the working of the virtual memory. d. Explain the Flynn's classification. e. Define: underflow and overflow.	CO4, L2 CO6, L1 CO4, L2 CO5, L2 CO3, L1	2x5=10
Q2	Explain the floating point representation with the help of example.	CO3, L2	10
Q3	Solve the expression: (-6) and (2) by using the booths multiplication method.	CO3, L3	10
Q4	Draw and explain the flow chart of arithmetic and instruction pipeline.	CO5, L2	5+5=10
Q5	Describe the cross bar and hypercube interconnection switch.	CO6, L2	5+5=10

Course Outcomes (CO)

Students will be able to

CO1: Demonstrate proficiency in register transfer language and perform operations including bus and memory transfers, as well as logic and shift micro-operations.

CO2: Assess various CPU architectures, I/O, memory organisation, addressing modes and modes of data transfer.

CO3: Apply computer arithmetic techniques effectively.

CO4: Study how the memory hierarchy influences system performance and focusing on the advance memories such as cache memory and virtual memory.

CO5: Analyze the parallel processing technique.

CO6: Understand the concept of multiprocessor, inter processor communication and synchronization.

RBT Classification	Lower Order Thinking Levels (LOTS)			Higher Order Thinking Levels (HOTS)		
RBT Level Number	L1	L2	L3	L4	L5	L6
RBT Level Name	Remembering	Understanding	Applying	Analyzing	Evaluating	Create